

can be denoted as a triple ordered set $(L, \oplus, \otimes)$.
 $(L, \wedge, \vee)$ is called algebraic lattice.

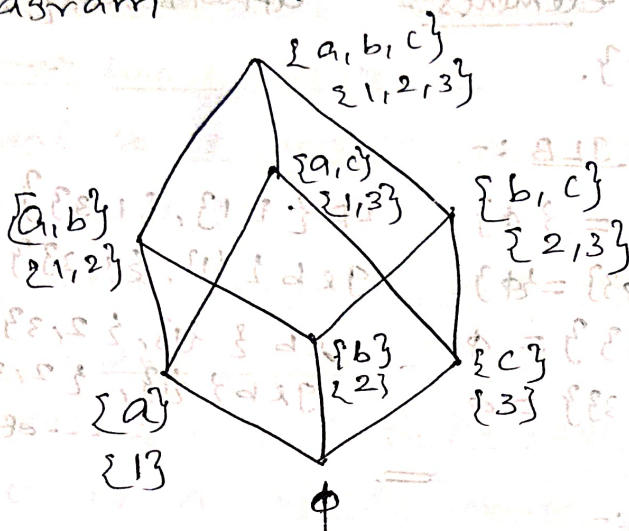
- ① Determine whether $(P(A), \subseteq)$, where $A = \{1, 2, 3\}$ is a lattice.

Sol: Given $A = \{1, 2, 3\}$.

Here $P(A)$ means power set.

$$P(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

Hasse Diagram



we see page no: 9.

Clearly L.U.B and G.L.B exists.

Hence $(P(A), \subseteq)$ is a lattice.

- ② Determine whether Posets represented by each of the Hasse diagram in figure are lattice.

